

A risk framework for addressing nature's harmful  
contributions to people in environmental  
governance

Matthew P. Faith<sup>1\*</sup>, Siân E. Rees<sup>1</sup>, Rory Gibb<sup>2</sup>,  
Briana Abrahms<sup>3</sup>, Lisa Hartmann<sup>4</sup>, Yvonne Walz<sup>4</sup>,  
Sally Janzen<sup>4</sup>, Holly J. Niner<sup>1</sup>, Bruno F. Simões<sup>5,6</sup>,  
Alexandre De Menezes<sup>7</sup>, Abigail McQuatters-Gollop<sup>1</sup>

<sup>1</sup>Marine Conservation Research Group, University of Plymouth, Drake  
Circus, Plymouth, PL4 8AA, UK.

<sup>2</sup>Centre for Biodiversity and Environment Research, Department of  
Genetics, Evolution and Environment, University College London,  
London, WC1E 6BT, UK.

<sup>3</sup>Center for Ecosystem Sentinels, Department of Biology, University of  
Washington, Seattle, WA, USA.

<sup>4</sup>Institute for Environment and Human Security, United Nations  
University, 1 Platz der Vereinten Nationen, 53113 Bonn, Germany.

<sup>5</sup>Rewilding Portugal, Quinta da Maunça s/n João Bravo, 6300-035  
Arrifana, Guarda, Portugal.

<sup>6</sup>School of Biological Sciences, Adelaide University, North Terrace, 5005  
Adelaide, South Australia, Australia.

<sup>7</sup>School of Biological and Chemical Sciences and Ryan Institute,  
Microbiology, Arts/Science Building, University of Galway, Republic of  
Ireland.

\*Corresponding author(s). E-mail(s): [matthew.faith@plymouth.ac.uk](mailto:matthew.faith@plymouth.ac.uk);

**Abstract**

Conserving Earth's ecosystems is critical for sustaining human life, yet some  
aspects of living nature, such as insect vectors of human disease and highly  
venomous animals, can harm people. Nature's harmful contributions to peo-  
ple (harmful NCPs) are a contentious issue in biodiversity research and policy,

with limited agreement on how they should be assessed or managed. Societal responses to harmful NCPs can undermine sustainability goals and result in ecosystem degradation, yet this risk is currently under-addressed in environmental governance. We therefore propose a new conceptual framework for assessing and managing harmful NCPs that draws parallels with the Intergovernmental Panel on Climate Change’s concept of climate risk. We argue that the risks of harmful NCPs should be understood as arising from the intersection of the hazardous aspects of ecosystems (‘ecosystem hazards’) with societal exposure and vulnerability. Our risk-based approach equips those decision makers responsible for environmental governance with a framework to systematically identify solutions for avoiding societal harms while preventing maladaptive interventions that can degrade nature. We conclude by outlining the required next-steps to operationalise our framework.

## Main

Earth’s ecosystems make invaluable contributions to people, from regulating the global climate system to providing food and medicinal resources [1]. Building on ecosystem service concepts, the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) refers to these ways living nature impacts human quality of life as Nature’s Contributions to People (NCPs; [1, 2]). Assessing the NCPs that benefit people can inform environmental governance that prioritises the protection and enhancement of human quality of life [3]. Some aspects of living nature, however, can reduce human quality of life [2, 4], and even directly harm people’s health, properties and livelihoods – hereafter referred to as ‘harmful NCPs’ (synonymous with material detrimental NCPs; see Box 1). Human vector-borne diseases, for example, cause more than 700,000 annual deaths [5] and severe economic burden [6], while venomous snakebites cause a reported 60,000 deaths annually [7, 8]. Crop pests and diseases, meanwhile, threaten food security and livelihoods, driving losses of up to 40% of global crop production and result in an annual cost of an estimated \$220 billion [9].

Harmful NCPs are an important consideration in environmental governance to ensure management decisions effectively protect people’s quality of life. While sustainable environmental governance largely focuses on regulating human pressures to safeguard beneficial NCPs [10], people’s quality of life could also be improved through regulating human pressures that amplify the intensity, frequency, and distribution of some harmful NCPs [11–13]. The potential to avoid harmful NCPs should also be considered during intergovernmental negotiations on greenhouse gas emissions, given increasing evidence identifies climate change as an amplifier of a range of pests, pathogens, and insect vectors of disease [14–17]. Certain harmful NCPs can be exacerbated by measures to protect and restore ecosystems, meaning they’re also an important consideration when planning ecosystem restoration initiatives. A relevant example of this is seen in increasing reintroductions of locally extinct species that

can amplify issues such as livestock depredation [18]. Finally, societal responses to harmful NCPs must be regulated to avoid ecosystem degradation and protect people from a loss of beneficial NCPs. Chemical pesticides, for example, are widely used in agriculture to control nuisance fungi, plants and insects [19], while large carnivores can be culled in attempts to reduce livestock depredation [20] and diseases [21]. Unsustainable levels of such interventions can degrade ecosystems [22–24] and threaten the long-term provision of beneficial NCPs.

Harmful NCPs have become a contentious issue in biodiversity research and policy [25, 26]. Although frameworks such as ‘ecosystem disservices’ [27] and ‘nature’s disvalues’ [4] have advanced the description and valuation of harmful NCPs, they have not been widely operationalised. In part, this is likely because such frameworks often only focus on the potential for specific species or habitats to cause societal harms without evaluating the likelihood of harms actually occurring. This approach frames governance decisions as an oversimplified trade-off between potential harms and benefits (i.e., the ecosystem service-disservice dichotomy) – overlooking whether harms can be avoided and inherently deprioritising the sustainable management of potentially-harmful species and habitats. We therefore propose a new conceptual framework for harmful NCPs that draws direct parallels with the concept of climate risk. The Intergovernmental Panel on Climate Change (IPCC) conceptualises climate risks as arising from the convergence of climate hazards with population exposure and vulnerability [28, 29]. While the hazard-exposure-vulnerability framework has been applied to zoonotic disease risks [30], we argue for applying it to a much broader suite of harmful NCPs. In this context, societal risks arise not from nature alone but from the intersection of the hazardous aspects of ecosystems (or ‘ecosystem hazards’) with societal exposure and vulnerability (Box 1). By reframing harmful NCPs as avoidable risks, our proposed approach identifies actionable intervention points that support risk-informed environmental governance decisions.

#### Box 1 - Glossary

##### **Nature’s beneficial contributions to people (beneficial NCPs):**

Aspects of living nature that increase human quality of life through material (e.g., food provision) and non-material (e.g., human-nature relationships) benefits. Living nature can also regulate material and non-material benefits (e.g., pollination that underpins food provision from cultivated crops or trees that stabilise soils and prevent landslides).

##### **Nature’s detrimental contributions to people (detrimental NCPs):**

Aspects of living nature that reduce human quality of life through material (e.g., zoonotic diseases) and non-material (e.g., wildlife phobias) harms. Living nature can also regulate material and non-material harms (e.g., habitats that support insect vectors of disease or plants that fuel wildfires).

**Nature’s harmful contributions to people (harmful NCPs):** Aspects of living nature that directly reduce human quality of life through material adverse impacts on human health, properties and livelihoods – synonymous with material detrimental NCPs.

**Ecosystem hazards:** Specific aspects of living nature that may cause tangible harms to human health, properties, and livelihoods (i.e., cause harmful NCPs).

**Exposure:** The physical contact between people (or their assets) and the ecosystem hazard from broad (e.g., spatial overlap between habitats and human-dominated environments) to fine (e.g., individual contact with organisms) scales.

**Vulnerability:** The potential for people (or their assets) to be adversely affected by the ecosystem hazard, determined by sensitivity and adaptive capacity (e.g., lack of preventative medicines or high socioeconomic dependency on exposed assets).

**Risk:** The likelihood of harms to human health, properties or livelihoods occurring, determined by the intersection of an ecosystem hazard with societal exposure and vulnerability.

**Ecosystem hazard mitigation:** measures to reduce the upstream drivers of ecosystem hazards, including through reducing hazard-amplifying human pressures (e.g., climate change and nutrient pollution) or not restoring potentially hazardous ecosystems (e.g., not reintroducing locally extinct carnivores to landscapes).

**Ecosystem hazard adaptation:** measures to manage the societal risks from actual or expected ecosystem hazards. Adaptation includes measures to reduce societal exposure and vulnerability (e.g., physical barriers and early warning systems), as well as measures that directly address hazardous species or habitats (e.g., wildlife population control and habitat clearances).

**Maladaptation:** measures intended to manage the societal risks from ecosystem hazards but instead exacerbate or transfer societal risks, particularly through measures that are ineffective at reducing risks from ecosystem hazards yet still drive ecosystem degradation and the loss of beneficial NCPs.

## An ecosystem hazard framework for harmful NCPs

Climate risks are complex and multifactorial, and so the IPCC’s risk framework signposts multiple potential avenues of intervention to reduce adverse societal impacts. The social-ecological systems that generate harmful NCPs are similarly

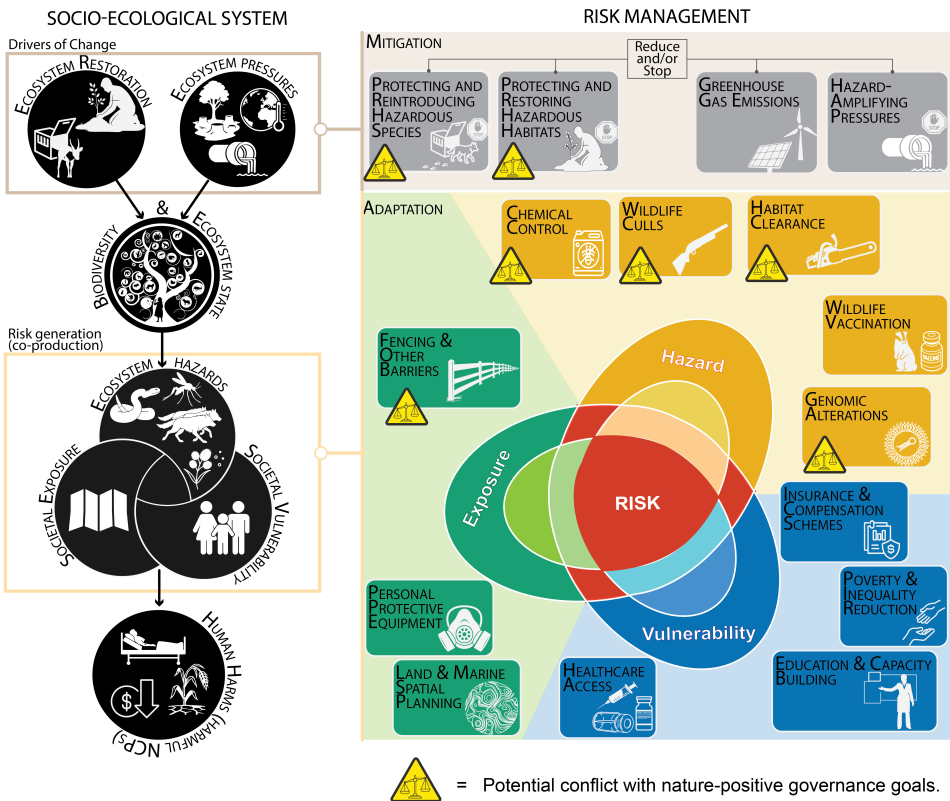
complex, where adverse societal impacts often result from people’s interactions with ecosystems that have potential to cause harm (i.e., NCP co-production; [31–33]). We term the specific threats in ecosystems that have the potential to harm people ‘ecosystem hazards’ and highlight that the hazard-exposure-vulnerability framework can untangle the complex social-ecological co-production dynamics that ultimately generate harmful NCPs (Box 1). Human population growth and subsequent land-use changes, for example, are global drivers of human exposure to hazardous species and habitats [34]. Likewise, a global expansion of aquaculture activities has been linked to increasing reports of harmful algal bloom impacts on shellfish aquaculture [35]. Socioeconomic factors can also determine the relative vulnerability of different communities to hazardous species and habitats; human mortalities from snakebites, for example, are primarily driven by limited access to antivenoms and healthcare rather than the regional diversity of venomous species [36, 37].

The IPCC approach to risk management distinguishes between mitigation and adaptation [28]. Mitigation addresses the upstream anthropogenic drivers of hazards, and so climate mitigation reduces the greenhouse gas emissions responsible for amplifying global climate hazards. Given certain ecosystem hazards can be similarly amplified by upstream anthropogenic drivers [11–13], such as human pressures on ecosystems, regulating these drivers can mitigate some ecosystem hazards (Box 1 and Figure 1). Adaptation efforts then serve to increase society’s resilience to actual or expected hazards, and so climate adaptation interventions reduce people’s exposure and vulnerability to climate hazards that persist despite mitigation measures [28]. Given harmful NCPs similarly result from societal exposure and vulnerability to ecosystem hazards, addressing patterns of exposure and vulnerability can reduce the risk of harmful NCPs being generated from ecosystem hazards (Figure 1). Importantly, measures that directly address specific hazards, such as restoring reefs to dissipate wave energy and reduce coastal flood risks under climate change, are classified as adaptation by the IPCC [38]. In the context of human-wildlife conflicts, however, mitigation is often used to broadly describe all risk-reduction measures, including measures to directly address hazardous wildlife, such as lethal control [39]. Adapting the climate risk framework for harmful NCPs therefore expands the terminology available to those decision makers responsible for sustainable environmental governance, and in doing so highlights multiple avenues for risk management.

## Mitigating ecosystem hazards

A major challenge for mitigating ecosystem hazards is the complex relationship between ecosystem changes and the distribution, frequency, and intensity of hazards. The availability of beneficial NCPs is generally dependent upon biodiversity and ecosystem functioning [2, 3, 10], meaning assessments of beneficial NCPs provide decision-makers with evidence of the potential societal harms stemming from ecosystem degradation. We cannot similarly generalise the relationship between ecosystem hazards and the state of ecosystems. Even the relationships between biodiversity loss

**Figure 1 - Applying the Intergovernmental Panel on Climate Change’s risk framework to ecosystem hazards highlights multiple risk management avenues for avoiding nature’s harmful contributions to people.**



The risk of harmful NCPs is conceptualised as the intersection of ecosystem hazards with societal exposure (spatiotemporal overlap with people) and vulnerability (societal sensitivity and overall adaptive capacity). This framework highlights diverse management opportunities for adapting to ecosystem hazards that range from spatial planning and physical barriers (reducing exposure) to healthcare access and compensation schemes (reducing vulnerability). Managing upstream drivers of ecosystem change (including through reducing greenhouse gas emissions) could also provide an opportunity to mitigate some ecosystem hazards. Some management measures, such as lethal control, can reduce risks of human harms from ecosystem hazards but can also undermine nature recovery targets and threaten the provision of nature’s beneficial contributions to people - examples of these interventions are marked as a trade-off.

and specific ecosystem hazards, such as the potential for zoonotic disease spillover into humans, have been hotly debated [40, 41]. There is increasing agreement, however, that some ecosystem hazards can increase in intensity, frequency or distribution under direct human pressures on ecosystems. Nutrient pollution, for example, can intensify some harmful algal blooms [13], while deforestation is known to drive some vector-borne diseases (e.g., ‘frontier’ and zoonotic malaria) [42, 43] and land-use changes can increase risks of zoonotic disease spillover [44]. Similarly, habitat fragmentation has been linked to increased frequency of wildlife raiding crops [11, 12]. Strategies to reduce anthropogenic pressures that amplify specific ecosystem hazards can thus

serve as ecosystem hazard mitigation (Box 1) and reduce associated societal risks.

Some ecosystem hazards are amplified by climate change. Given many NCPs are generated by specific species (i.e., Species' or Wildlife's Contributions to People; [45, 46]), climate-driven changes to hazardous species can impact the intensity, frequency and distribution of some ecosystem hazards (Table 1). Warmer temperatures, for example, are driving upward trends and range expansions of certain crop diseases [17], harmful algal blooms [47], environmental pathogens [16, 48], and vector-borne diseases [15, 49–51]. The impacts of sudden-onset climate events on certain species can also drive some ecosystem hazards; droughts, for example, can deplete the wild food resources available to wildlife, driving increased livestock depredation and crop raiding [52]. Other NCPs, however, are generated by broader ecosystem processes [46], such as carbon sequestration that is driven by complex interactions between different species [53]. Climate change is also impacting such ecosystem processes, including those that have the potential to generate harmful NCPs. Prolonged and more intense pollen dispersal under warming temperatures, for example, are intensifying seasonal allergies in some countries [54], while increasingly intense freshwater harmful algal blooms have been linked to climate-driven changes in lake hydrology [55]. International progress on mitigating climate change effects through reducing greenhouse gas emissions could therefore limit the redistribution, frequency, and intensity of some climate-amplified ecosystem hazards and should be considered a component of ecosystem hazard mitigation (Box 1).

**Table 1: Specific ecosystem hazards amplified by anthropogenic climate change's impact on hazardous species.** Selected examples of organisms associated with specific ecosystem hazards and their responses to climate change. Following the IPCC risk framework adopted here, climate change can act as a risk amplifier through intensifying ecosystem hazards and thus increasing direct risks to human health, property, and livelihoods.

| Organism                                                    | Hazardous trait                                                                                        | Hazard response to climate change                                                                                                                                                                                                                        |
|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Coccidioides</i> spp. (fungus)                           | Human pathogen which causes Valley Fever.                                                              | Observed increased Valley Fever rates in the US as increasing droughts and dust storms alter <i>Coccidioides</i> spp. range and spore dispersal [16].                                                                                                    |
| <i>Aedes aegypti</i> and <i>A. albopictus</i> (mosquitoes)  | Vector of human pathogens including dengue, chikungunya, and zika.                                     | Projected redistribution of disease driven by shifts in species' range, alongside changes in incidence in existing endemic areas (disease transmission rates are thermally-regulated), with changing air temperature, rainfall and humidity [49–51, 56]. |
| <i>Gambierdiscus</i> spp. (microalgae)                      | Produces neurotoxins that cause Ciguatera Fish Poisoning in humans who consume contaminated reef fish. | Increasing ocean temperatures are projected to redistribute Ciguatera poisoning risks, with increased growth rates in parts of <i>Gambierdiscus</i> ' existing geographical range and a range expansion into temperate latitudes [47].                   |
| <i>Aspergillus flavus</i> and <i>A. parasiticus</i> (fungi) | Produces aflatoxins (a human carcinogen), which contaminates crops.                                    | An observed increase in crop contamination with aflatoxins in temperate latitudes with increasing temperatures [57].                                                                                                                                     |

*Continued on next page...*

Table 1: (continued)

| Organism                                             | Hazardous trait                                                                     | Hazard response to climate change                                                                                                                                                                                                                       |
|------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Melinis minutiflora</i><br>(molasses grass)       | Highly flammable, increasing the potential for wildfires.                           | Projected expansion of wildfire likelihood to higher elevations as warming temperatures make previously cool alpine areas suitable for invasion and redistribute the accumulation of flammable biomass [58].                                            |
| <i>Carcharodon carcharias</i><br>(great white shark) | Large predator which can bite humans.                                               | An observed increase in shark bite incidences during strong El Niño events, which are projected to increase in severity, as high sea surface temperatures drive sharks to shift their distribution nearshore and overlap with people [52, 59].          |
| <i>Loxodonta africana</i><br>(African bush elephant) | Large herbivore which can destroy crops and water infrastructure and injure humans. | Observed increase in crop raiding and presence in human settlements during severe droughts. As climate change drives increasingly frequent and severe droughts, such conflicts are projected to intensify due to escalating resource scarcity [52, 60]. |
| <i>Ursus maritimus</i><br>(polar bear)               | Large predator which can attack humans.                                             | An observed increase in human-polar bear interactions as warming drives long-term sea ice decline, forcing bears to spend more time on land and into human-dominated environments [52, 61].                                                             |
| <i>Locusta migratoria</i><br>(migratory locust)      | Crop pest which can form swarms and cause substantial crop losses.                  | Projected redistribution of crop damages as warming facilitates range expansion, while increased tropical cyclones and heavy rainfall favour swarm formation [14, 62].                                                                                  |

Mitigating ecosystem hazards through curbing drivers of ecosystem degradation, such as nutrient pollution and climate-warming greenhouse gas emissions, can simultaneously protect people from harmful NCPs and losses of beneficial NCPs [10, 63]. Reducing ecosystem pressures can come with economic costs [64], so understanding where these interventions can also protect society from harmful NCPs provides valuable justification for governance decisions. Some ecosystem hazards, however, can be mitigated through limiting the protection, restoration, or reintroduction of hazardous species and habitats. Limiting the restoration of wetlands, for example, can prevent increases in the populations of insect vectors of disease [65], while large carnivores are often not reintroduced or protected in order to avoid livestock losses [66]. However, mitigating ecosystem hazards through broader reluctance to protect and restore ecosystems conflicts with nature-positive governance goals (Figure 1) and reduces the potential for beneficial NCPs generation. Adaptation measures therefore provide an alternative avenue for risk reduction.

## Adapting to ecosystem hazards

International progress towards nature recovery and climate targets is slow. Although the rate of increase in annual global greenhouse gas emissions may have started to decline [67], major challenges stand in the way of reaching net zero emissions [68],

with historical emissions having ‘baked in’ a substantial amount of future climate change (and associated impacts; [69]). Similarly, examples of successes in meeting international nature recovery targets at a country or regional scale are overshadowed by a continuing global trend of biodiversity declines and ecosystem degradation [70, 71]. Consequently, like climate hazards, society must adapt to pressure-driven ecosystem hazards which could continue to increase in intensity, frequency and distribution despite mitigation efforts. In addition, ecosystem hazard adaptation can build societal resilience to hazards that are naturally present in biodiverse and functioning ecosystems. Adaptation measures can thus help communities avoid harm when protecting and restoring hazardous species and habitats - reducing trade-offs between nature recovery and human quality of life, and helping restoration initiatives to deliver a net benefit to people (through beneficial NCPs). Ecosystem hazard adaptation can be categorised into three approaches: managing hazardous species and habitats, managing societal exposure, and managing societal vulnerability.

**Managing ecosystem hazards** addresses the direct management of existing species and habitats that have the potential to harm people (Box 1). While mitigation measures reduce the upstream drivers of ecosystem hazards, ecosystem hazards can also be directly reduced through interventions such as wildlife population control and habitat clearances [39]. However, such measures can often conflict with nature recovery targets (Figure 1), creating a trade-off between protecting society from immediate harm and minimising ecosystem degradation that erodes beneficial NCPs. Agricultural pesticide usage, for instance, can increase immediate crop yields but also place pressure on non-target organisms [19, 22], such as pollinators that underpin food security [72]. Human-wildlife conflicts capture similar trade-offs, where the immediate threat of crop-raiding elephants or livestock-depredating carnivores can drive lethal control efforts at the cost of nature-positive governance goals [52]. Despite these trade-offs, some hazard-reducing measures can align with nature recovery targets, such as reducing populations of hazardous species that are invasive. Eucalyptus trees introduced to southern Europe, for example, can elevate wildfire likelihood [73] and negatively affect biodiversity and ecosystem functioning [74] - reducing their extent can thus support nature recovery whilst simultaneously reducing societal risks.

**Managing societal exposure** addresses the spatiotemporal overlap between people (and their assets) and ecosystem hazards (Box 1). At the fine scale, implementing physical barriers and deterrents is a popular strategy for deterring hazardous wildlife: increasing protection of cattle at night or using guardian animals can reduce livestock exposure to large carnivores [75], while mosquito nets and other household-level controls limit human contact with insect vectors of disease [30, 76]. Many exposure-reducing management measures can be responsive to temporary increases in ecosystem hazard likelihood. When harmful algal blooms develop, for example, societal harms can be avoided through temporary closures of aquaculture sites and beaches [77], while proactive beach closures following shark sightings are an effective strategy for avoiding shark bites [78]. Land-use and Marine Spatial Planning can also proactively avoid human harms by strategic avoidance of hazard hotspots – for

instance, siting new aquaculture facilities away from areas prone to recurrent harmful algal blooms [79].

417

418 **Managing societal vulnerability** addresses the sensitivity of people (and their  
419 assets) to ecosystem hazards, alongside overall adaptive capacity (Box 1). Reducing  
420 people’s sensitivity to ecosystem hazards that impact health can include strategies  
421 such as vaccination programmes that build immunity to zoonotic diseases [30].  
422 Addressing the sensitivity of people’s livelihoods and properties to ecosystem hazards  
423 is a further component of risk management. Livestock vaccination, for example, has  
424 been shown to reduce the economic sensitivity of farmers to livestock losses from  
425 predators [80], while diversifying crop planting regimes in agricultural systems can  
426 reduce the sensitivity of plantations to pests [81]. Building adaptive capacity is also an  
427 important component of managing vulnerability to ecosystem hazards, such as devel-  
428 oping early warning systems for harmful algal blooms that allow aquaculture sites  
429 to harvest stock before site closures (or relocate stock to avoid impacts all together;  
430 [82]). Ecosystem hazard adaptive capacity can be limited by economic resources (Box  
431 2), so efforts to address global inequalities (e.g., advancing the Sustainable Develop-  
432 ment Goals; [83]) can support communities to adapt to ecosystem hazards.

433

434

435 **Box 2 – Socioeconomic inequality as a driver of vulnerability to**  
436 **ecosystem hazards.**

437

438 Like climate-related disasters [84], socioeconomic inequalities can drive dis-  
439 proportionate societal risks from ecosystem hazards. People who rely on rivers  
440 for water and bathing, for example, face heightened exposure to crocodiles [52],  
441 while many zoonotic diseases are associated with exposure inherent to pas-  
442 toral livelihoods, hunting, or the wildlife trade [30]. More broadly, deaths from  
443 snakebites are more strongly driven by access to antivenoms and proximity to  
444 medical care than the regional diversity of venomous snake species [36]. This  
445 pattern extends to emerging zoonotic diseases, where healthcare access is a  
446 strong predictor of reported outbreaks across a wide diversity of pathogens [85],  
447 and global vaccine inequity amplifies human health impacts from vector-borne  
448 diseases [86]. Similarly, livestock depredation causes disproportionate financial  
449 harms to farmers in low- and middle-income countries [87]. Those communities  
450 that are disproportionately exposed and vulnerable to ecosystem hazards are  
451 often those that depend most on Nature’s beneficial Contributions to People,  
452 such as the provision of fresh water as drinking resource [63]. At the same  
453 time, these communities often must rely on responses to ecosystem hazards  
454 that degrade ecosystems and threaten the provision of beneficial NCPs, such as  
455 unsustainable levels of lethal control or habitat removal. Such short-term protec-  
456 tive responses arise from the immediate need to protect health and livelihoods,  
457 and from limited options to build long-term adaptive capacities.

458

459

460

Capacity building to address inequalities in exposure and vulnerability to ecosystem hazards is therefore a human development priority [88], to safeguard people from direct harms and prevent long-term losses of beneficial NCPs. Such capacity building should ensure adequate evidence is available to support regional ecosystem hazard risk management. Global biodiversity databases are heavily skewed towards nations with greater resources [89], limiting the availability to accurately map and assess ecosystem hazards in data-poor regions. In addition, research on human-wildlife conflict mitigation is predominantly conducted in North America, Europe, and Australia – often overlooking the specific socio-economic and ecological contexts of the Global South [39, 90]. Investment in research capacity can therefore generate valuable evidence to guide effective risk management. The FAIR (Findable, Accessible, Interoperable, Reusable) and CARE (Collective benefit, Authority to control, Responsibility, Ethics) principles [91] are then vital to ensure the sovereignty of any generated data is respected [92]. Capacity building can also include education (e.g., snakebite education programmes) that empowers communities to understand and navigate the risks they face [93, 94], alongside larger-scale investments such as building healthcare capacity or developing early warning systems [95, 96].

Adaptation measures that address societal exposure and vulnerability generally present less conflicts with nature-positive governance goals (Figure 1) and so are often advocated as human-wildlife coexistence measures. Guidelines published by the International Union for the Conservation of Nature and the Species Survival Commission [75], for example, highlight measures such as using reinforced enclosures for livestock at night to prevent carnivore predation (exposure reduction), and compensation schemes to offset livelihood or economic losses (vulnerability reduction). However, these adaptation measures can also present trade-offs; fencing, for example, is widely employed to protect livestock from predators but overuse can drive habitat fragmentation and adversely impact ecosystems [97]. In addition, an increasing suite of tools directly addresses hazardous species and habitats while minimising biodiversity loss. Wildlife vaccination initiatives, for example, can reduce livestock disease risks and be an effective alternative to wildlife culls [98, 99]. Similarly, in contrast to mosquito population suppression strategies [100], newer gene-editing approaches seek to engineer malaria resistance in mosquitoes [101], preserving their ecological roles [102] while reducing impacts on human health. Systematically assessing all avenues for risk reduction is thus essential to identifying the most sustainable adaption options for specific ecosystem hazards.

Importantly, some societal responses to ecosystem hazards can degrade ecosystems and reduce beneficial NCPs yet still fail to avoid harmful NCPs. Culling wild badger populations in the United Kingdom, for example, did not curb tuberculosis transmission to livestock [21], while culling vampire bats in Peru appears to have amplified, rather than reduced, rabies transmission [103, 104]. Within climate risk management, adaptation strategies that transfer or exacerbate risks are considered maladaptive

[105]. By analogy, ecosystem hazard management should also be considered maladaptive (Box 1) when it is ineffective at reducing risks and simultaneously drives ecosystem degradation – thereby eroding beneficial NCPs and creating additional threats to society.

511

## 512 **Towards operationalising ecosystem hazard risk** 513 **assessments** 514

515 To effectively operationalise our ecosystem hazard framework for harmful NCPs, we  
516 identify four priorities:  
517

518 **1. Categorising ecosystem hazards.** Operationalising risk assessments will  
519 require metrics to consistently categorise and measure ecosystem hazards. IPBES  
520 defines a suite of generalisable NCP variables that includes categories such as food  
521 and feed provision [1]. These variables can act as standardised indicators to link  
522 ecological changes with potential societal impacts, helping to prioritise governance  
523 interventions [63, 106]. Some of the existing NCP variables used by IPBES are linked  
524 to harmful NCPs. The creation and maintenance of habitats [1], for example, can support  
525 both species that generate beneficial NCPs but also those that generate harmful  
526 NCPs. However, direct human harms, such as human injuries and infectious diseases  
527 caused by wildlife, are not currently included in the IPBES NCP framework. To  
528 address this gap, established hazard classifications, such as the United Nations Office  
529 for Disaster Risk Reduction and International Science Council’s Hazard Information  
530 Profiles [107], should be used to create standardised ecosystem hazard indicators  
531 from ecological datasets.  
532

533 **2. Ecosystem hazard attribution.** Ecosystem hazard mitigation depends  
534 upon understanding the dynamics between specific human pressures and the intensity,  
535 frequency and distribution of ecosystem hazards. Attribution methods attempt  
536 to quantify the human contribution to biodiversity changes [108], yet the inherent  
537 complexities of ecosystems make these quantitative approaches challenging [109].  
538 While we advocate for the development of ecosystem hazard attribution tools to  
539 systematically prioritise where ecosystem hazard mitigation could effectively reduce  
540 societal risks, complementary evidence syntheses are also required to broadly understand  
541 the role of human pressures in ecosystem hazard generation. The One Health  
542 paradigm has been particularly successful in motivating rigorous evidence syntheses  
543 focussed on the role of ecosystem health in disease emergence [110]. Broadening such  
544 evidence syntheses to cover the full diversity of ecosystem hazards could therefore  
545 underpin efforts to mitigate ecosystem hazards through regulating specific ecosystem  
546 pressures. Importantly, evidence syntheses on the drivers of ecosystem hazards should  
547 incorporate local knowledge that can provide valuable insights of historical ecological  
548 changes, their drivers, and human consequences [111, 112].  
549

550 **3. Operationalising risk assessments across governance scales.** Broad-scale  
551 indicators that combine hazard, exposure, and vulnerability metrics are valuable for  
552

identifying and projecting global climate risk hotspots [29, 113]. Developing similar indicators for ecosystem hazard risks could identify priority regions for investing in ecosystem hazard adaptive capacity building measures (Box 1) and project the future distribution of risks from climate-amplified ecosystem hazards. However, global-scale risk indicators often overlook the regional ecosystem dynamics and local community needs that determine whether environmental governance interventions are locally supported, just, and effective [114]. Regional governance mechanisms should therefore follow a bottom-up approach to building local-level resilience to ecosystem hazards that adopts community-led governance processes [115, 116]. Importantly, the stakeholders who face the greatest risks from an ecosystem hazard (e.g., farmers facing livestock losses) can differ from those accessing the beneficial NCPs from the same species or habitat (e.g., ecotourism providers; [117]). Participatory mapping can therefore help to navigate different community perspectives and increase local support for governance interventions [118]. Local knowledge of ecosystems can also provide valuable evidence for locally effective risk-management, such as knowledge of predator movements that can guide herding practices to avoid livestock depredation [119]. Co-produced [120, 121] regional ecosystem hazard risk assessments should therefore ensure local knowledge of ecosystems is captured to maximise the effectiveness of risk management.

**4. Addressing non-material detrimental NCPs.** Our ecosystem hazard framework addresses NCPs that cause material harm to human health, properties and livelihoods (Box 1), yet living nature can also be perceived as detrimental to people without causing tangible harm [4]. Certain habitats, for example, might be perceived by some people as unsightly, while certain species may hold negative cultural associations [122]. Addressing non-material detrimental NCPs is therefore vital to ensure environmental governance decisions are culturally appropriate and locally supported. However, the hazard-exposure-vulnerability framework typically addresses risks stemming from biophysical hazards, and so addressing non-material detrimental NCPs requires a different approach to risk assessments that quantify hazard, exposure, and vulnerability (i.e., index-based assessments). As a complement to these quantitative methods, the IPCC generally adopt qualitative, narrative-based approaches to assess the non-material impacts of climate change climate, such as documenting losses of cultural heritage and psychological trauma caused by extreme weather events [123, 124]. We similarly advocate for qualitative, narrative-based approaches to addressing the non-material societal impacts caused by living nature that might arise from environmental governance interventions. Given negative perceptions of different species and habitats are highly context-specific and vary between different cultures (and even individuals), qualitative assessment approaches are likely to be most useful for guiding regional-level environmental governance decisions.

## 599 Summary

600 Nature’s harmful contributions to people (harmful NCPs) are a natural component  
601 of Earth’s ecosystems, yet societal responses often result in ecosystem degradation  
602 that threatens the future provision of beneficial NCPs. Here, we adapt the IPCC’s  
603 climate risk framework to reframe harmful NCPs as avoidable risks, arising from the  
604 intersection of ecosystem hazards with societal exposure and vulnerability. Opera-  
605 tionalising our framework will require standardised ecosystem hazard classifications,  
606 attribution frameworks and evidence syntheses to identify where regulating upstream  
607 human drivers could mitigate hazards, and risk indicators to prioritise risk manage-  
608 ment interventions. To ensure risk management is fair, just, and supported by the  
609 people who interact with hazardous species and habitats, risk assessments should be  
610 co-produced with local communities and address subjective negative perceptions of  
611 ecosystems. Advancing our ecosystem hazard framework will support environmen-  
612 tal governance to protect people’s quality of life whilst navigating the sustainability  
613 trade-offs of risk management interventions.  
614

615  
616 **Acknowledgements.** MPF acknowledges funding from the UK Department for  
617 Environment, Food and Rural Affairs (DEFRA), the University of Plymouth, and The  
618 Alan Turing Institute Enrichment Scheme. RG is supported by UCL and the Wolfson  
619 Foundation (via a Wolfson UCL Excellence Fellowship) and the Royal Society (via a  
620 University Research Fellowship). Views and opinions expressed are, however, those of  
621 the authors only and do not necessarily reflect those of the funders.  
622

623 **Conflict of interest.** None declared.

624 **Author contributions.** Wrote the paper: MPF, SER, RG, BA, LH, YW, SJ, HJN,  
625 BFS, ADM, and AMG.  
626

## 627 628 References 629

- 630 [1] Brondizio, E. S., Settele, J., Diaz, S. & Ngo, H. T. (eds) *Global Assess-*  
631 *ment Report on Biodiversity and Ecosystem Services of the Intergovernmental*  
632 *Science-Policy Platform on Biodiversity and Ecosystem Services* (IPBES secre-  
633 tariat, Bonn, Germany, 2019).  
634  
635 [2] Díaz, S. *et al.* Assessing nature’s contributions to people. *Science* **359**, 270–272  
636 (2018).  
637  
638 [3] Brauman, K. A. *et al.* Global trends in nature’s contributions to people.  
639 *Proceedings of the National Academy of Sciences* **117**, 32799–32805 (2020).  
640  
641 [4] Lliso, B., Lenzi, D., Muraca, B., Chan, K. M. & Pascual, U. Nature’s disvalues:  
642 what are they and why do they matter? *Current Opinion in Environmental*  
643 *Sustainability* **56**, 101173 (2022).  
644

- [5] World Health Organization. Vector-borne diseases (2024). URL <https://www.who.int/news-room/fact-sheets/detail/vector-borne-diseases>. Accessed: 2025-01-10.
- [6] Shepard, D. S., Undurraga, E. A., Halasa, Y. A. & Stanaway, J. D. The global economic burden of dengue: a systematic analysis. *The Lancet infectious diseases* **16**, 935–941 (2016).
- [7] Collaborators', G. . S. E. Global mortality of snakebite envenoming between 1990 and 2019. *Nature Communications* **13**, 6160 (2022).
- [8] Farooq, H. *et al.* Snakebite incidence in rural sub-saharan africa might be severely underestimated. *Toxicon* **219**, 106932 (2022).
- [9] Food and Agriculture Organization of the United Nations. The hidden health crisis: How plant diseases threaten global food security (2025). URL <https://www.fao.org/one-health/highlights/how-plant-diseases-threaten-global-food-security>. Accessed: 2026-01-10.
- [10] Cimatti, M., Chaplin-Kramer, R. & Di Marco, M. The role of high-biodiversity regions in preserving nature's contributions to people. *Nature Sustainability* **6**, 1385–1393 (2023).
- [11] Wettasin, M. *et al.* Environmental factors induced crop raiding by wild asian elephant (*elephas maximus*) in the eastern economic corridor, thailand. *Scientific Reports* **13**, 13388 (2023).
- [12] Montero-Botey, M., Kivuyo, E., Sitati, N. & Perea, R. Deforestation and water availability as main drivers of human-elephant conflict. *Global Ecology and Conservation* **54**, e03068 (2024).
- [13] Davidson, K. *et al.* Anthropogenic nutrients and harmful algae in coastal waters. *Journal of environmental management* **146**, 206–216 (2014).
- [14] Ma, C.-S. *et al.* Crop pest responses to global changes in climate and land management. *Nature Reviews Earth Environment* **6**, 264–283 (2025).
- [15] Carlson, C. J. *et al.* Pathogens and planetary change. *Nature Reviews Biodiversity* **1**, 32–49 (2025).
- [16] Howard, M., Sayes, C., Giesy, J. P. & Li, Y. Valley fever under a changing climate in the united states. *Environment International* 109066 (2024).
- [17] Singh, B. K. *et al.* Climate change impacts on plant pathogens, food security and paths forward. *Nature Reviews Microbiology* **21**, 640–656 (2023).
- [18] Strand, G.-H., Hansen, I., de Boon, A. & Sandström, C. Carnivore management zones and their impact on sheep farming in norway. *Environmental management*

691           **64**, 537–552 (2019).  
692  
693 [19] Tang, F. H., Lenzen, M., McBratney, A. & Maggi, F. Risk of pesticide pollution  
694 at the global scale. *Nature geoscience* **14**, 206–210 (2021).  
695  
696 [20] Merz, L. M., von Hohenberg, B. C., Bergmann, N. T., Bruskotter, J. T. &  
697 Carter, N. H. Elusive effects of legalized wolf hunting on human-wolf interactions.  
698 *Science Advances* **11**, eadu8945 (2025).  
699  
700 [21] Langton, T. E., Jones, M. W. & McGill, I. Analysis of the impact of bad-  
701 ger culling on bovine tuberculosis in cattle in the high-risk area of england,  
702 2009–2020. *Veterinary Record* **190**, e1384 (2022).  
703  
704 [22] Wan, N.-F. *et al.* Pesticides have negative effects on non-target organisms.  
705 *Nature communications* **16**, 1360 (2025).  
706  
707 [23] Ripple, W. J. *et al.* Status and ecological effects of the world’s largest carnivores.  
708 *Science* **343**, 1241484 (2014).  
709  
710 [24] Vincenot, C. E., Florens, F. V. & Kingston, T. Can we protect island flying  
711 foxes? *Science* **355**, 1368–1370 (2017).  
712  
713 [25] Villa, F. *et al.* The misconception of ecosystem disservices: How a catchy term  
714 may yield the wrong messages for science and society. *Ecosystem Services* **10**,  
715 52–53 (2014).  
716  
717 [26] Blanco, J., Dendoncker, N., Barnaud, C. & Sirami, C. Ecosystem disservices  
718 matter: Towards their systematic integration within ecosystem service research  
719 and policy. *Ecosystem services* **36**, 100913 (2019).  
720  
721 [27] Shackleton, C. M. *et al.* Unpacking pandora’s box: Understanding and categoris-  
722 ing ecosystem disservices for environmental management and human wellbeing.  
723 *Ecosystems* **19**, 587–600 (2016).  
724  
725 [28] Möller, V. *et al.* in *Annex II: Glossary* (eds Pörtner, H.-O. *et al.*) *Climate*  
726 *Change 2022: Impacts, Adaptation and Vulnerability. Contribution of Working*  
727 *Group II to the Sixth Assessment Report of the Intergovernmental Panel on*  
728 *Climate Change* 2897–2930 (Cambridge University Press, Cambridge, UK and  
729 New York, NY, USA, 2022).  
730  
731 [29] Alegría, A., Poloczanska, E., Loeschke, S., Mintenbeck, K. & Poertner, H.  
732 Towards an ipcc atlas for comprehensive climate change risk assessments. *npj*  
733 *Climate Action* **3**, 108 (2024).  
734  
735 [30] Gibb, R., Redding, D. W., Friant, S. & Jones, K. E. Towards a ‘people  
736 and nature’paradigm for biodiversity and infectious disease. *Philosophical*  
*Transactions B* **380**, 20230259 (2025).

- [31] Fischer, A. & Eastwood, A. Coproduction of ecosystem services as human–nature interactions—an analytical framework. *Land use policy* **52**, 41–50 (2016). 737  
738  
739
- [32] Blanco, J. *et al.* A novel ecosystem (dis) service cascade model to navigate sustainability problems and its application in a changing agricultural landscape in brazil. *Sustainability Science* **17**, 105–119 (2022). 740  
741  
742  
743
- [33] Kachler, J., Isaac, R., Martín-López, B., Bonn, A. & Felipe-Lucia, M. R. Co-production of nature’s contributions to people: What evidence is out there? *People and Nature* **5**, 1119–1134 (2023). 744  
745  
746  
747
- [34] Ma, D. *et al.* Global expansion of human-wildlife overlap in the 21st century. *Science Advances* **10**, eadp7706 (2024). 748  
749  
750
- [35] Hallegraeff, G. M. *et al.* Perceived global increase in algal blooms is attributable to intensified monitoring and emerging bloom impacts. *Communications Earth Environment* **2**, 117 (2021). 751  
752  
753  
754
- [36] Longbottom, J. *et al.* Vulnerability to snakebite envenoming: a global mapping of hotspots. *The Lancet* **392**, 673–684 (2018). 755  
756  
757
- [37] Berg, P., Theart, F., van Driel, M., Saaiman, E. L. & Mavoungou, L.-B. Snakebite envenoming in africa remains widely neglected and demands multidisciplinary attention. *nature communications* **15**, 9598 (2024). 758  
759  
760
- [38] Parmesan, C. *et al.* in *Terrestrial and freshwater ecosystems and their services* (eds Pörtner, H.-O. *et al.*) *Climate Change 2022: Impacts, Adaptation and Vulnerability* 197–377 (Cambridge University Press, Cambridge, UK and New York, NY, USA, 2022). 761  
762  
763  
764  
765
- [39] Lazure, L. & Weladji, R. B. Methods to mitigate human–wildlife conflicts involving common mesopredators: a meta-analysis. *The Journal of Wildlife Management* **88**, e22526 (2024). 766  
767  
768  
769
- [40] Rohr, J. R. *et al.* Towards common ground in the biodiversity–disease debate. *Nature ecology evolution* **4**, 24–33 (2020). 770  
771  
772
- [41] Keesing, F. & Ostfeld, R. S. Impacts of biodiversity and biodiversity loss on zoonotic diseases. *Proceedings of the National Academy of Sciences* **118**, e2023540118 (2021). 773  
774  
775  
776
- [42] De Castro, M. C., Monte-Mor, R. L., Sawyer, D. O. & Singer, B. H. Malaria risk on the amazon frontier. *Proceedings of the National Academy of Sciences* **103**, 2452–2457 (2006). 777  
778  
779  
780
- [43] Brock, P. M. *et al.* Predictive analysis across spatial scales links zoonotic malaria to deforestation. *Proceedings of the Royal Society B* **286**, 20182351 (2019). 781  
782

783 [44] Gibb, R. *et al.* Zoonotic host diversity increases in human-dominated ecosys-  
784 tems. *Nature* **584**, 398–402 (2020).  
785  
786 [45] Chaplin-Kramer, R. *et al.* Wildlife’s contributions to people. *Nature Reviews*  
787 *Biodiversity* **1**, 68–81 (2025).  
788  
789 [46] Rey, P.-L., Adde, A., Külling, N., Lehmann, A. & Guisan, A. Advancing species-  
790 based predictions of nature’s contributions to people. *Trends in Ecology &*  
791 *Evolution* (2026).  
792  
793 [47] Tester, P. A., Litaker, R. W. & Berdalet, E. Climate change and harmful benthic  
794 microalgae. *Harmful Algae* **91**, 101655 (2020).  
795  
796 [48] Bekele, B. K. *et al.* Cholera in africa: A climate change crisis. *Journal of*  
797 *epidemiology and global health* **15**, 68 (2025).  
798  
799 [49] Farooq, Z. *et al.* Impact of climate and aedes albopictus establishment on dengue  
800 and chikungunya outbreaks in europe: a time-to-event analysis. *The Lancet*  
801 *Planetary Health* **9**, e374–e383 (2025).  
802  
803 [50] Childs, M. L., Lyberger, K., Harris, M., Burke, M. & Mordecai, E. A. Climate  
804 warming is expanding dengue burden in the americas and asia. *medRxiv* 2024.01.  
805 08.24301015 (2025).  
806  
807 [51] Gibb, R. *et al.* Interactions between climate change, urban infrastructure and  
808 mobility are driving dengue emergence in vietnam. *Nature communications* **14**,  
809 8179 (2023).  
810  
811 [52] Abrahms, B. *et al.* Climate change as a global amplifier of human–wildlife  
812 conflict. *Nature Climate Change* **13**, 224–234 (2023).  
813  
814 [53] Soto-Navarro, C. *et al.* Mapping co-benefits for carbon storage and biodiversity  
815 to inform conservation policy and action. *Philosophical Transactions of the*  
816 *Royal Society B: Biological Sciences* **375**, 20190128 (2020). URL <https://doi.org/10.1098/rstb.2019.0128>.  
817  
818 [54] Anderegg, W. R. *et al.* Anthropogenic climate change is worsening north amer-  
819 ican pollen seasons. *Proceedings of the National Academy of Sciences* **118**,  
820 e2013284118 (2021).  
821  
822 [55] Ho, J. C., Michalak, A. M. & Pahlevan, N. Widespread global increase in intense  
823 lake phytoplankton blooms since the 1980s. *Nature* **574**, 667–670 (2019).  
824  
825 [56] Ryan, S. J., Carlson, C. J., Mordecai, E. A. & Johnson, L. R. Global expansion  
826 and redistribution of aedes-borne virus transmission risk with climate change.  
827 *PLoS neglected tropical diseases* **13**, e0007213 (2019).  
828

- [57] Bunny, S. M., Umar, A., Bhatti, H. S. & Honey, S. F. Aflatoxin risk in the era of climatic change-a comprehensive review. *CABI Agriculture and Bioscience* **5**, 105 (2024). 829  
830  
831  
832
- [58] Maciel, J. C. *et al.* The distribution potential of melinis minutiflora (molasses grass) under current and future climates for europe using climex tool. *Theoretical and Applied Climatology* **156**, 353 (2025). 833  
834  
835  
836
- [59] Chapman, B. K. & McPhee, D. Global shark attack hotspots: Identifying underlying factors behind increased unprovoked shark bite incidence. *Ocean Coastal Management* **133**, 72–84 (2016). 837  
838  
839  
840
- [60] Mariki, S. B., Svarstad, H. & Benjaminsen, T. A. Elephants over the cliff: explaining wildlife killings in tanzania. *Land Use Policy* **44**, 19–30 (2015). 841  
842  
843
- [61] Towns, L., Derocher, A., Stirling, I., Lunn, N. & Hedman, D. Spatial and temporal patterns of problem polar bears in churchill, manitoba. *Polar Biology* **32**, 1529–1537 (2009). 844  
845  
846
- [62] Liu, X., Zhang, D. & He, X. Unveiling the role of climate in spatially synchronized locust outbreak risks. *Science Advances* **10**, eadj1164 (2024). 847  
848  
849
- [63] Chaplin-Kramer, R. *et al.* Global modeling of nature’s contributions to people. *Science* **366**, 255–258 (2019). 850  
851  
852
- [64] Waldron, A. *et al.* Protecting 30% of the planet for nature: costs, benefits and economic implications (2020). 853  
854  
855
- [65] Dworrak, T. V., Sauer, F. G. & Kiel, E. Wetland conservation and its effects on mosquito populations. *Wetlands* **42**, 96 (2022). 856  
857  
858
- [66] Treves, A. & Bruskotter, J. Tolerance for predatory wildlife. *Science* **344**, 476–477 (2014). 859  
860  
861
- [67] Friedlingstein, P. *et al.* Global carbon budget 2023. *Earth System Science Data* **15**, 5301–5369 (2023). 862  
863  
864
- [68] Muttitt, G., Price, J., Pye, S. & Welsby, D. Socio-political feasibility of coal power phase-out and its role in mitigation pathways. *Nature Climate Change* **13**, 140–147 (2023). 865  
866  
867
- [69] Solomon, S., Plattner, G.-K., Knutti, R. & Friedlingstein, P. Irreversible climate change due to carbon dioxide emissions. *Proceedings of the national academy of sciences* **106**, 1704–1709 (2009). 868  
869  
870  
871
- [70] Díaz, S. *et al.* Pervasive human-driven decline of life on earth points to the need for transformative change. *Science* **366**, eaax3100 (2019). 872  
873  
874

875 [71] Maney, C. *et al.* National commitments to aichi targets and their implica-  
876 tions for monitoring the kunming-montreal global biodiversity framework. *npj*  
877 *Biodiversity* **3**, 6 (2024).  
878  
879 [72] Woodcock, B. A. *et al.* Impacts of neonicotinoid use on long-term population  
880 changes in wild bees in england. *Nature communications* **7**, 12459 (2016).  
881  
882 [73] Bousfield, C. G. *et al.* Global risk of wildfire across timber production systems.  
883 *Nature Communications* **16**, 4204 (2025).  
884  
885 [74] Iglesias-Carrasco, M. *et al.* Global impacts of exotic eucalypt plantations on  
886 wildlife. *Biological Reviews* (2025).  
887  
888 [75] IUCN SSC. *IUCN SSC Guidelines on Human-Wildlife Conflict and Coexistence*  
889 (International Union for Conservation of Nature, Gland, Switzerland, 2023).  
890  
891 [76] Pryce, J., Richardson, M. & Lengeler, C. Insecticide-treated nets for preventing  
892 malaria. *Cochrane Database of Systematic Reviews* (2018).  
893  
894 [77] Karlson, B. *et al.* Harmful algal blooms and their effects in coastal seas of  
895 northern europe. *Harmful Algae* **102**, 101989 (2021).  
896  
897 [78] Huveneers, C. *et al.* Shifts in the incidence of shark bites and efficacy of beach-  
898 focussed mitigation in australia. *Marine pollution bulletin* **198**, 115855 (2024).  
899  
900 [79] Stoner, O., Economou, T., Torres, R., Ashton, I. & Brown, A. R. Quantifying  
901 spatio-temporal risk of harmful algal blooms and their impacts on bivalve shell-  
902 fish mariculture using a data-driven modelling approach. *Harmful Algae* **121**,  
903 102363 (2023).  
904  
905 [80] Young, J. C. *et al.* Collaborative conservation for snow leopards: Lessons  
906 learned from successful community-based interventions. *Conservation Letters*  
907 **17**, e13062 (2024).  
908  
909 [81] Altieri, M. A., Nicholls, C. I., Dinelli, G. & Negri, L. Towards an agroecologi-  
910 cal approach to crop health: reducing pest incidence through synergies between  
911 plant diversity and soil microbial ecology. *npj Sustainable Agriculture* **2**, 6  
912 (2024).  
913  
914 [82] Davidson, K. *et al.* Habreports: online early warning of harmful algal and  
915 biotoxin risk for the scottish shellfish and finfish aquaculture industries. *Frontiers*  
916 *in Marine Science* **8**, 631732 (2021).  
917  
918 [83] United Nations. Transforming our world: The 2030 agenda for sustainable  
919 development. Tech. Rep., United Nations (2015). URL [https://sdgs.un.org/](https://sdgs.un.org/2030agenda)  
920 [2030agenda](https://sdgs.un.org/2030agenda).

|      |                                                                                                 |     |
|------|-------------------------------------------------------------------------------------------------|-----|
| [84] | Raju, E., Boyd, E. & Otto, F. Stop blaming the climate for disasters.                           | 921 |
|      | <i>Communications Earth Environment</i> <b>3</b> , 1 (2022).                                    | 922 |
|      |                                                                                                 | 923 |
| [85] | Gibb, R. <i>et al.</i> The anthropogenic fingerprint on emerging infectious diseases.           | 924 |
|      | <i>MedRxiv</i> 2024.05. 22.24307684 (2024).                                                     | 925 |
|      |                                                                                                 | 926 |
| [86] | Zhang, K. <i>et al.</i> Impact of climate change on vaccine responses and inequity.             | 927 |
|      | <i>Nature Climate Change</i> 1–3 (2024).                                                        | 928 |
|      |                                                                                                 | 929 |
| [87] | Braczkowski, A. R. <i>et al.</i> The unequal burden of human-wildlife conflict.                 | 930 |
|      | <i>Communications Biology</i> <b>6</b> , 182 (2023).                                            | 931 |
|      |                                                                                                 | 932 |
| [88] | Amorim-Maia, A. T. & Olazabal, M. Beyond adjustment: a new paradigm for                         | 933 |
|      | climate change adaptation in a complex world. <i>Global Environmental Change</i>                | 934 |
|      | <b>93</b> , 103027 (2025).                                                                      | 935 |
|      |                                                                                                 | 936 |
| [89] | García-Roselló, E., González-Dacosta, J. & Lobo, J. M. The biased distribution                  | 937 |
|      | of existing information on biodiversity hinders its use in conservation, and we                 | 938 |
|      | need an integrative approach to act urgently. <i>Biological Conservation</i> <b>283</b> ,       | 939 |
|      | 110118 (2023).                                                                                  | 940 |
|      |                                                                                                 | 941 |
| [90] | Newsom, A., Sebesvari, Z. & Dorresteyn, I. Climate change influences the risk                   | 942 |
|      | of physically harmful human-wildlife interactions. <i>Biological Conservation</i> <b>286</b> ,  | 943 |
|      | 110255 (2023).                                                                                  | 944 |
|      |                                                                                                 | 945 |
| [91] | Carroll, S. R., Herczog, E., Hudson, M., Russell, K. & Stall, S. Operationalizing               | 946 |
|      | the care and fair principles for indigenous data futures. <i>Scientific data</i> <b>8</b> , 108 | 947 |
|      | (2021).                                                                                         | 948 |
|      |                                                                                                 | 949 |
| [92] | Walter, M. <i>et al.</i> Indigenous data sovereignty in the era of big data and open            | 950 |
|      | data. <i>Australian Journal of Social Issues</i> <b>56</b> , 143–156 (2021).                    | 951 |
|      |                                                                                                 | 952 |
| [93] | Vaiyapuri, S. <i>et al.</i> Multifaceted community health education programs as pow-            | 953 |
|      | erful tools to mitigate snakebite-induced deaths, disabilities, and socioeconomic               | 954 |
|      | burden. <i>Toxicon: X</i> <b>17</b> , 100147 (2023).                                            | 955 |
|      |                                                                                                 | 956 |
| [94] | Petriello, M. A. <i>et al.</i> The scope of empowerment for conservation and                    | 957 |
|      | communities. <i>Conservation Biology</i> <b>39</b> , e14249 (2025).                             | 958 |
|      |                                                                                                 | 959 |
| [95] | Williams, D. J. <i>et al.</i> Strategy for a globally coordinated response to a prior-          | 960 |
|      | ity neglected tropical disease: Snakebite envenoming. <i>PLoS neglected tropical</i>            | 961 |
|      | <i>diseases</i> <b>13</b> , e0007059 (2019).                                                    | 962 |
|      |                                                                                                 | 963 |
| [96] | Carvajal-Yepes, M. <i>et al.</i> A global surveillance system for crop diseases. <i>Science</i> | 964 |
|      | <b>364</b> , 1237–1239 (2019).                                                                  | 965 |
|      |                                                                                                 | 966 |
| [97] | Sun, J. <i>et al.</i> Reconsidering the efficiency of grazing exclusion using fences on         |     |
|      | the tibetan plateau. <i>Science Bulletin</i> <b>65</b> , 1405–1414 (2020).                      |     |

- 967 [98] Martin, S. *et al.* Is moving from targeted culling to bcg-vaccination of bad-  
968 gers (meles meles) associated with an unacceptable increased incidence of cattle  
969 herd tuberculosis in the republic of ireland? a practical non-inferiority wildlife  
970 intervention study in the republic of ireland (2011-2017). *Preventive veterinary*  
971 *medicine* **179**, 105004 (2020).  
972  
973 [99] Freuling, C. M. *et al.* The elimination of fox rabies from europe: determinants  
974 of success and lessons for the future. *Philosophical Transactions of the Royal*  
975 *Society B: Biological Sciences* **368**, 20120142 (2013).  
976  
977 [100] Kyrou, K. *et al.* A crispr-cas9 gene drive targeting doublesex causes com-  
978 plete population suppression in caged anopheles gambiae mosquitoes. *Nature*  
979 *biotechnology* **36**, 1062–1066 (2018).  
980  
981 [101] Carballar-Lejarazú, R. *et al.* Dual effector population modification gene-drive  
982 strains of the african malaria mosquitoes, anopheles gambiae and anopheles  
983 coluzzii. *Proceedings of the National Academy of Sciences* **120**, e2221118120  
984 (2023).  
985  
986 [102] Lahondère, C. *et al.* The olfactory basis of orchid pollination by mosquitoes.  
987 *Proceedings of the National Academy of Sciences* **117**, 708–716 (2020).  
988  
989 [103] Viana, M. *et al.* Effects of culling vampire bats on the spatial spread and spillover  
990 of rabies virus. *Science Advances* **9**, eadd7437 (2023).  
991  
992 [104] Miguel, E. *et al.* A systemic approach to assess the potential and risks of wildlife  
993 culling for infectious disease control. *Communications Biology* **3**, 353 (2020).  
994  
995 [105] Higuera Roa, O., Walz, Y. & Nehren, U. How to avoid the risk of maladaptation?  
996 from a conceptual understanding to a systematic approach for analyzing poten-  
997 tial adverse effects in adaptation actions. *Mitigation and Adaptation Strategies*  
998 *for Global Change* **30**, 27 (2025).  
999  
1000 [106] Chaplin-Kramer, R. *et al.* Mapping the planet’s critical natural assets. *Nature*  
1001 *Ecology & Evolution* **7**, 51–61 (2023).  
1002  
1003 [107] United Nations Office for Disaster Risk Reduction & International Science Coun-  
1004 cil. UNDRR-ISC Hazard Information Profiles - 2025 Update. Report, United  
1005 Nations Office for Disaster Risk Reduction and International Science Council  
1006 (2025).  
1007  
1008 [108] Gonzalez, A., Chase, J. M. & O’Connor, M. I. A framework for the detection  
1009 and attribution of biodiversity change. *Philosophical Transactions of the Royal*  
1010 *Society B* **378**, 20220182 (2023).  
1011  
1012 [109] Schrodtt, F. *et al.* Advancing causal inference in ecology: Pathways for biodi-  
1013 versity change detection and attribution. *Methods in ecology and evolution* **16**,

|                                                                                                  |      |
|--------------------------------------------------------------------------------------------------|------|
| 2276–2304 (2025).                                                                                | 1013 |
|                                                                                                  | 1014 |
| [110] Daszak, P. <i>et al.</i> Workshop report on biodiversity and pandemics of the intergov-    | 1015 |
| ernmental platform on biodiversity and ecosystem services. Tech. Rep., IPBES                     | 1016 |
| secretariat, Bonn, Germany (2020).                                                               | 1017 |
|                                                                                                  | 1018 |
| [111] Savo, V. <i>et al.</i> Observations of climate change among subsistence-oriented           | 1019 |
| communities around the world. <i>Nature Climate Change</i> <b>6</b> , 462–473 (2016).            | 1020 |
|                                                                                                  | 1021 |
| [112] Giovos, I. <i>et al.</i> Integrating local ecological knowledge, citizen science and long- | 1022 |
| term historical data for endangered species conservation: Additional records of                  | 1023 |
| angel sharks (chondrichthyes: Squatinidae) in the mediterranean sea. <i>Aquatic</i>              | 1024 |
| <i>Conservation: Marine and Freshwater Ecosystems</i> <b>29</b> , 881–890 (2019).                | 1025 |
|                                                                                                  | 1026 |
| [113] Faith, M. P. <i>et al.</i> Mapping global fisheries climate risks to guide sustainable     | 1027 |
| marine management. <i>bioRxiv</i> 2025.05. 07.652593 (2025).                                     | 1028 |
|                                                                                                  | 1029 |
| [114] Eriksen, S. <i>et al.</i> Adaptation interventions and their effect on vulnerability in    | 1030 |
| developing countries: Help, hindrance or irrelevance? <i>World development</i> <b>141</b> ,      | 1031 |
| 105383 (2021).                                                                                   | 1032 |
|                                                                                                  | 1033 |
| [115] Farhad, S. <i>et al.</i> Ecosystem services governance: A cross-realm lever for            | 1034 |
| sustainability transformation. <i>Ecosystem Services</i> <b>74</b> , 101748 (2025).              | 1035 |
|                                                                                                  | 1036 |
| [116] Brondizio, E. S. & Tourneau, F.-M. L. Environmental governance for all. <i>Science</i>     | 1037 |
| <b>352</b> , 1272–1273 (2016).                                                                   | 1038 |
|                                                                                                  | 1039 |
| [117] Jani, V. Human–wildlife conflict or human–human conflict? social constructions             | 1040 |
| of stakeholder groups involved in wildlife management in northern zimbabwe.                      | 1041 |
| <i>Integrative Conservation</i> <b>4</b> , 218–230 (2025).                                       | 1042 |
|                                                                                                  | 1043 |
| [118] Brown, G. & Raymond, C. M. Methods for identifying land use conflict potential             | 1044 |
| using participatory mapping. <i>Landscape and Urban Planning</i> <b>122</b> , 196–208            | 1045 |
| (2014).                                                                                          | 1046 |
|                                                                                                  | 1047 |
| [119] Ivaşcu, C. M. & Biro, A. Coexistence through the ages: the role of native                  | 1048 |
| livestock guardian dogs and traditional ecological knowledge as key resources                    | 1049 |
| in conflict mitigation between pastoralists and large carnivores in the romanian                 | 1050 |
| carpathians. <i>Journal of Ethnobiology</i> <b>40</b> , 465–482 (2020).                          | 1051 |
|                                                                                                  | 1052 |
| [120] Chambers, J. M. <i>et al.</i> Six modes of co-production for sustainability. <i>Nature</i> | 1053 |
| <i>sustainability</i> <b>4</b> , 983–996 (2021).                                                 | 1054 |
|                                                                                                  | 1055 |
| [121] Satterthwaite, E. V. <i>et al.</i> Centering knowledge co-production in sustainability     | 1056 |
| science. <i>Oceanography</i> <b>37</b> , 26–37 (2024).                                           | 1057 |
|                                                                                                  | 1058 |
| [122] Carucci, T. <i>et al.</i> Ecosystem services and disservices associated with vultures:     |      |
| A systematic review and evidence assessment. <i>Ecosystem Services</i> <b>56</b> , 101447        |      |

1059 (2022).  
1060  
1061 [123] Tschakert, P., Ellis, N. R., Anderson, C., Kelly, A. & Obeng, J. One thousand  
1062 ways to experience loss: A systematic analysis of climate-related intangible harm  
1063 from around the world. *Global Environmental Change* **55**, 58–72 (2019).  
1064  
1065 [124] Waiwai, M. *et al.* Case study on non-economic loss & damage to Vanuatu’s  
1066 coastal ecosystems and community livelihoods from slow onset events to support  
1067 the design and operationalization of the Loss & Damage Fund (2023). URL  
1068 [https://unfccc.int/sites/default/files/resource/Vanuatu%20TC%20Workshop%](https://unfccc.int/sites/default/files/resource/Vanuatu%20TC%20Workshop%20Case%20Study%20on%20NELD%20from%20SOEs%2025%20April%202023.pdf)  
1069 [20Case%20Study%20on%20NELD%20from%20SOEs%2025%20April%202023.](https://unfccc.int/sites/default/files/resource/Vanuatu%20TC%20Workshop%20Case%20Study%20on%20NELD%20from%20SOEs%2025%20April%202023.pdf)  
1070 [pdf](https://unfccc.int/sites/default/files/resource/Vanuatu%20TC%20Workshop%20Case%20Study%20on%20NELD%20from%20SOEs%2025%20April%202023.pdf).  
1071  
1072  
1073  
1074  
1075  
1076  
1077  
1078  
1079  
1080  
1081  
1082  
1083  
1084  
1085  
1086  
1087  
1088  
1089  
1090  
1091  
1092  
1093  
1094  
1095  
1096  
1097  
1098  
1099  
1100  
1101  
1102  
1103  
1104